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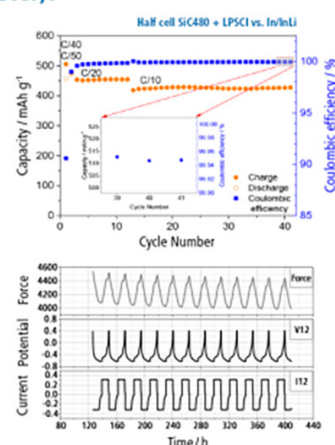
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Experimental sol-gel fabrication and theoretical simulation of one-dimensional photonic crystals with a defect state

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Abstract. One-dimensional multilayer structures of the periodically alternating low refractive index (silica) and high refractive index (titania) materials have been deposited by sol-gel spin-coating. The possibility of creating an optical filter by introducing a structural defect in the form of a layer of double thickness is shown theoretically and experimentally. All experimental spectra are in agreement with theoretical spectra calculated by transfer matrix method.

1. Introduction

Photonic band gap structures (PBGs), also known as photonic crystals (PCs), are structures whose refractive index is periodic on a length scale of the order of optical wavelengths preventing light propagating through the structure due to Bragg reflection. Depending on the dimensionality of this periodicity, PBGs can be one, two or three dimensions (1D, 2D or 3D). In the case of sol-gel derived materials, the primary emphasis has been on 1D- and 3D-structures [1]. Distributed Bragg reflectors (DBR), also called Bragg mirrors, are examples of 1D-PBGs.

Sol-gel technology is one of the simplest and cheapest technologies for the synthesis of photonic crystals. Sol-gel DBRs can be made, for example, by dip- [2] or spin-coating [3] of colloidal nanoparticles. Silica (SiO_2) and titania (TiO_2) are usually chosen because of the high optical contrast of their refractive indices (~ 1.46 and ~ 2.4 , respectively).

In optics, filters often need narrowband transmission of photons in a narrow range of wavelengths in units of nanometers wide. To obtain such a narrow-band filter structure PCs some defects can be introduced: 1) the layers with double (triple, quadruple, etc.) or some aliquant thicknesses, 2) the layers of different material.

2. Experimental section

An experimental technique for the synthesis of multilayer structure was spin-coating of silica and titania sols on a glass substrate, followed by annealing of layers in a muffle furnace. The technique in details has been described previously [4].

The transfer matrix method was used to calculate multilayer structure transmission spectra [5]. Matrix A describes light propagation through the layers boundary, and matrix B describes light propagation through current material layer:



$$A = \frac{1}{2n_2} \begin{pmatrix} n_1 + n_2 & n_2 - n_1 \\ n_2 - n_1 & n_1 + n_2 \end{pmatrix},$$

$$B = \begin{pmatrix} \exp\left(2\pi i \frac{nL}{\lambda}\right) & 0 \\ 0 & \exp\left(-2\pi i \frac{nL}{\lambda}\right) \end{pmatrix},$$

where n_1 and n_2 are refractive indices of alternating materials, n and L – refractive index and thickness of the current layer respectively, λ – the light wavelength.

The dispersion curves of the refractive indices $n_1(\lambda)$ and $n_2(\lambda)$ used in the calculations respectively for SiO_2 [6] and TiO_2 [7] are following:

$$n_1 = \sqrt{1 + \frac{0.6962\lambda^2}{\lambda^2 - 0.0684^2} + \frac{0.4079\lambda^2}{\lambda^2 - 0.1162^2} + \frac{0.8975\lambda^2}{\lambda^2 - 9.9961^2}},$$

$$n_2 = \sqrt{5.913 + \frac{0.2441}{\lambda^2 - 0.0803}}.$$

The described matrices are multiplied for all layers of the structure and all layers boundaries. The result of the multiplication is the transfer matrix M . Elements of alternate matrix multiplication M ($M = A_1 B_1 A_2 B_2 \dots A_n B_n$) allows to describe the optical properties of the multilayer structure. This common method of spectrum simulation can be used not only for photonic crystals but also for any multilayer structures including aperiodic ones.

To investigate producing of photonic crystals filters by sol-gel technology it has been researched 1D-PC structures with different numbers and thicknesses of layers. The structural defect was achieved by doubling the layer of titanium dioxide in the center of the structure.

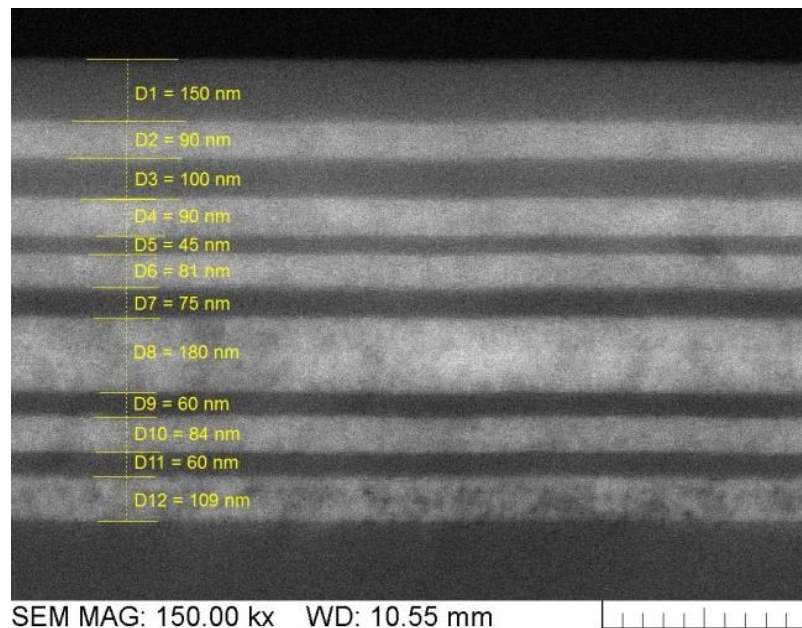


Figure 1. Scanning electron microscopy (secondary electrons) of photonic crystals with thicknesses of alternating layers (bar is 500 nm)

3. Results and discussion

The PC structure with defect has been characterized by scanning electron microscopy (Fig. 1). The light bands correspond to titanium dioxide, and dark ones are for silica. Therefore, thinner silicon dioxide stripes interspersed with thicker layers of silicon oxide. The average thicknesses are 70 and 100 nm respectively.

Theoretical and experimental transmission spectra of the photonic crystal with double titania defect layer are compared in Fig. 2a. The position of the defect energy level inside the stop band and the quality factor of the photonic crystal is different in theory and experiment. The quality factor difference is explained due to ideal wall approximation in the transfer matrix model while there are non-zero porosity and roughness of layers in the experiment. When the porosity is taken into account, the maximum associated with a defect state shifts to shorter wavelengths because of the lower refractive index of titania and silica films in Fig.2a.

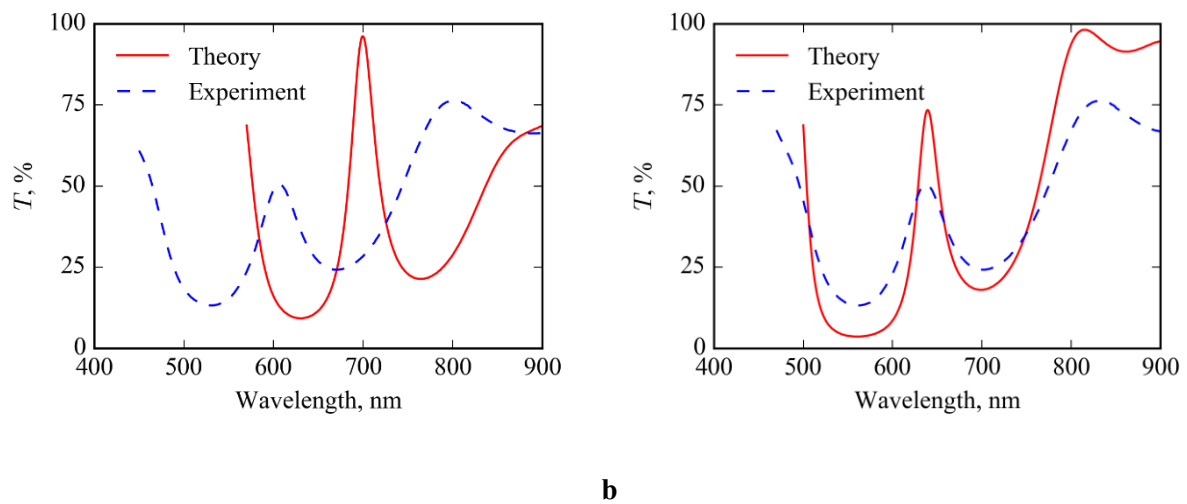


Figure 2. Theoretical and experimental transmission spectra of the photonic crystal with defect double titania layer in the center of the structure (a) in the case of continuous material dispersion curve, (b) in the case of refractive indices taking into account porosity of films.

The porous structure of the thin films composing the photonic crystal can be taken into account by modifying the expressions for the refractive indices $n_1(\lambda)$ and $n_2(\lambda)$.

$$n_1 = \sqrt{0.5 + \frac{0.6962\lambda^2}{\lambda^2 - 0.0684^2} + \frac{0.4079\lambda^2}{\lambda^2 - 0.1162^2} + \frac{0.8975\lambda^2}{\lambda^2 - 9.9961^2}},$$

$$n_2 = \sqrt{4.913 + \frac{0.2441}{\lambda^2 - 0.0803}}.$$

The proposed correction allows one to obtain the wavelength of the defect state corresponding to the experiment (Fig. 2b). However, this approach does not take into account the roughness of the layers, which leads to an experimentally observed decrease in the quality factor. Also, it is worth noting that the difference in layers is also due to the technique of structure synthesis – different times of the layers in the muffle furnace lead to small differences in the dispersion curves of the layers for the same material.

4. Conclusions

Although sol-gel technology is one of the cheapest methods for the production of one-dimensional photonic crystals, using this technique and simulating photonic crystals, one must take into account the porosity of the layers. Since annealing time affects variate refractive index values, instead of ellipsometric measurements the simulation of the inverse problem can be used for reconstructing refractive indices from transmission spectra with known layer thicknesses.

Acknowledgments

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